

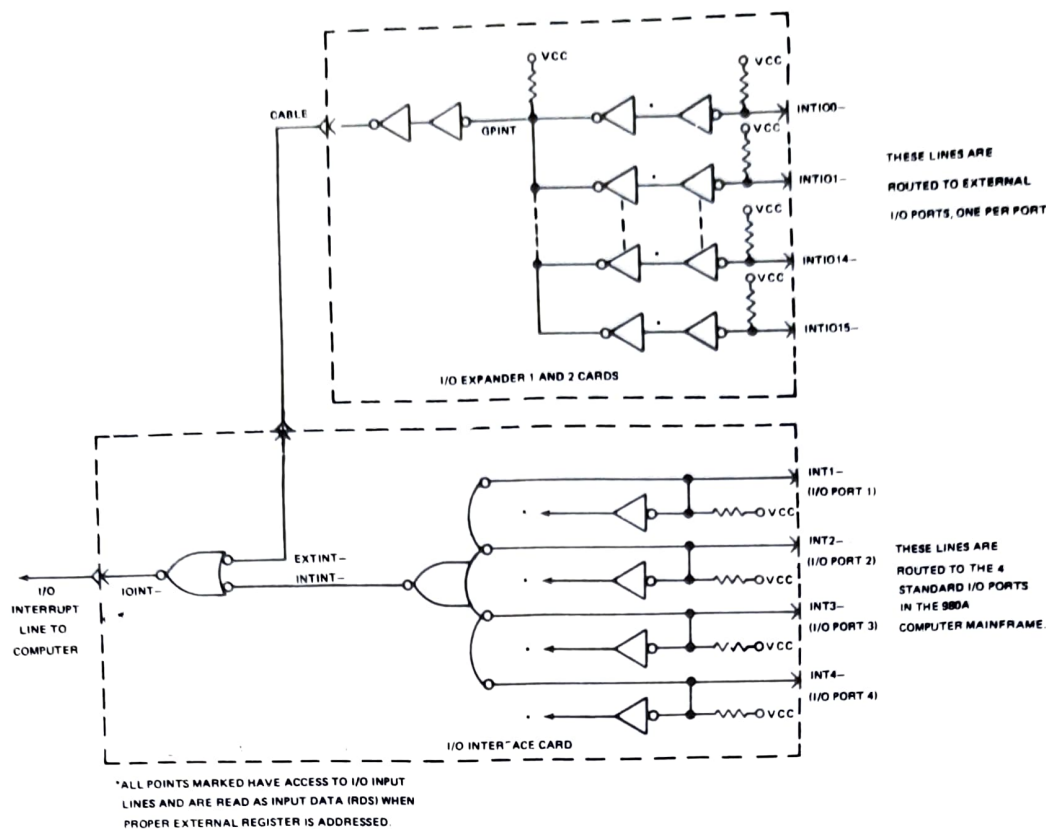


FIGURE 2-3. INPUT/OUTPUT INTERRUPTS TO CENTRAL PROCESSING UNIT

2-2.1 READ DIRECT SINGLE INSTRUCTION. The Read Direct Single (RDS) instruction consists of two 16-bit words (WD1 and WD2) as illustrated in Figure 2-4. The first word of the RDS instruction (WD1) is applied to the output bus and an output strobe (G0) is generated by the ER field that its data must enter the input bus. This data is retained on the input bus until a release pulse (TERM) is generated by the Model 980A.

The destination of the input data is controlled by the second word of the instruction (WD2). The format of the read direct instruction is illustrated in Figure 2-4 and the control functions are described in separate subparagraphs which follow.

2-2.1.1 External Register Number (ER and Group). The ER and group specifies a particular device and/or device function. This 6 bit field (bits 9 and 11 through 15) permits the use of a maximum of 64 devices and/or functions. Bits 5 and 6 permit the use of four groups of 64 devices.

2-2.1.2 Busy Bit (B). This field provides an optional feature for use with devices that may not be ready to transfer data when queried by the CPU. If this bit is a logic

ONE, and a successful data transfer is performed, the next sequential instruction is skipped. If this bit is a logic ONE and no data transfer takes place, the next sequential instruction is executed. If this bit is a ZERO, data transfer will occur unconditionally and the next instruction in sequence is executed.

WORD 1

0	4	5	6	7	8	9	10	11	15	
1	1	0	1	1	GROUP	0	0	ER	0	ER

WORD 2

0	7	8	9	10	11	12	13	14	15
UNUSED				B	0	I	D	A	R

FIGURE 2-4. READ DIRECT SINGLE INSTRUCTION WORD FORMAT

2-2.1.3 Internal Register Number (R). The "R" field specifies which internal register is to be used in this instruction.

2-2.1.4 Address Mode Field (A). If the "A" field contains a logic ZERO, data is transferred to the register specified by the "R" field. If the "A" field contains a logic ONE, data is transferred to memory at the location specified by the contents of the register specified by the "R" field.

2-2.1.5 Decrement (D). When the "D" field is a logic ONE and the "A" field is a logic ONE, the address in the register specified by the "R" field is decremented by one each time a data word is transferred.

2-2.1.6 Increment (I). When the "I" field is a logic ONE and the "A" field is a logic ONE, the address in the register specified by the "R" field is incremented by one each time a data word is transferred. If "I" and "D" bits are both a logic ONE, the "I" bit will override. If the "I" bit is a logic ZERO, a decrement is executed.

2-2.2 WRITE DIRECT SINGLE INSTRUCTION. The Write Direct Single (WDS) instruction consists of two 15-bit words (WD1 and WD2) as illustrated in Figure 2-5. The WDS instruction is similar to the RDS instruction except for the direction of data flow, which is from the Model 980A to the external device.

2-3 EXPANSION.

Bus, interrupt, and group expansion are described in detail in separate subparagraphs which follow.

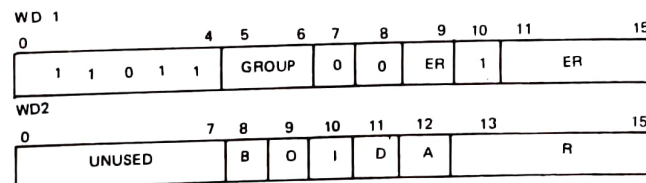


FIGURE 2-5. WRITE DIRECT SINGLE INSTRUCTION WORD FORMAT

2-3.1 BUS EXPANSION. Since there are 4 groups of 64 possible external register numbers available to address the I/O controllers, up to 256 (4X64) I/O devices may be connected to the Model 980A through the I/O structure. Input/Output expansion is accomplished internal to the Model 980A mainframe by use of an optional expansion backplane (TI Part No. 960703-0001), or externally by use of a separate I/O expander chassis (TI Part No. 216150) that is designed for installation in a standard equipment rack. Two additional cards (I/O expander cards 1 and 2) are utilized when the I/O bus is expanded. Figure 2-6 illustrates the I/O interface card, I/O expansion cable, and the two I/O expander cards.

2-3.1.1 Internal Expansion. When expansion is accomplished internally, I/O expander 1 and I/O expander 2 are installed in the optional expansion backplane (TI Part No. 960744-0001) of the Model 980A (See Figure 2-7). A cable (TI Part No. 217059-0002) from the I/O interface board is connected to I/O expander number 1 to complete the expansion signal interface.

The optional expansion backplane contains 12 ports which can be wired to accommodate any combination of I/O, DMAC, and API devices.

The standard configuration of the optional expansion backplane (TI Part No. 960703-0001) is wired to permit 9 of the 12 ports to be used for I/O devices. Two of the remaining ports (EX1 and EX2) are required for I/O expanders 1 and 2, respectively. The remaining one port (REG) is reserved for the optional ± 15 volt regulator card. The addition of the 9 optional I/O ports provides a total of 13 I/O ports available within the Model 980A mainframe.

2-3.1.2 External Expansion. When expansion is accomplished externally to the Model 980A mainframe, I/O expander 1 and I/O expander 2 are installed in the separate I/O expander chassis. A cable (TI Part No. 217059-XXXX) from the I/O interface board to I/O expander 1 completes the expansion signal interface (Figure 2-8).

The I/O expander chassis contains two backplanes. Either one or both of the backplanes can be configured to accept I/O expander cards 1 and 2, and provide 16 additional I/O ports.

2-3.1.3 Multiple Expansion. For large I/O systems, more than one expansion backplane may be required. Up to four additional I/O interface cards may be installed in the basic I/O ports. Each I/O interface card is cabled to another expansion backplane. Also, the I/O interface card may be

2-4 EXTERNAL REGISTER ASSIGNMENTS.

Each I/O controller has one or more external register numbers which are standard, although each controller is capable of being wired for any external register number. A list of standard external register number assignments for controllers operated through the I/O structure is provided in Figure 2-9 and Table 2-2.

2-5 THEORY OF OPERATION.

This paragraph contains a brief theoretical description of the I/O instructions, the I/O interface card and the I/O expander cards. A functional diagram of the I/O bus is illustrated in Figure 2-10.

2-5.1 ARITHMETIC UNIT CONTROL. The Arithmetic Unit (AU) in the Model 980A Computer contains control and data path logic for the I/O bus. When an RDS or WDS instruction is acquired, the AU ROM controller gates the first word of the instruction on the I/O output data lines for 750 nanoseconds or three clock periods (250 nanoseconds). During the second clock period a GO signal is applied to the I/O bus which signifies that the I/O controllers should decode the external register lines to determine which controller or function is being addressed by the I/O instruction. The RDS and WDS timing is illustrated in Figure 2-11 and 2-12, respectively.

The top two waveforms in Figures 2-11 and 2-12 include the RDS/WDS Word 1 and GO signal timing. If the "SKIP ON READY" option is used (RDS/WDS Word 2, bit 8 = ONE), the I/O controller should supply a READY signal to the AU within 1.5 microseconds as shown in Figures 2-11 and 2-12. The AU then produces a TERM signal to indicate acceptance of input data for the RDS instruction (Figure 2-11) or to indicate that output data is available to the I/O controller for a WDS instruction (Figure 2-12).

2-5.2 INPUT/OUTPUT INTERFACE. The I/O interface card which normally occupies card slot IO0 in the Model 980A mainframe provides a means to expand the I/O bus to other backplanes. The I/O interface card also contains the logic necessary to OR the interrupt lines from the four I/O ports in the Model 980A basic mainframe to form one I/O interrupt line to the CPU. Jumper wires enable the user to select the address to read the I/O interrupts as described in Paragraph 2-3.

A detailed diagram of the I/O interface logic is illustrated in Figure 2-13. Two types of symbols should be noted. When a connector pin symbol ($\rightarrow$) is shown, the connector is labeled P1 or P2. This label is followed by a dash and the pin number. The P1 designation indicates the connector is a bottom connector and P2 designates a top connector which can complete the connection to a cable with the proper cable connector.

The jumper pad symbol ($\rightarrow$) indicates a pad that is provided to permit jumper wires to be inserted for address selection. Normally, the pad identifier (E19 for example), as shown is also reproduced on the component side of the printed circuit board.

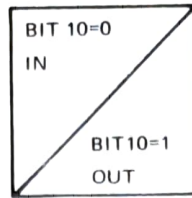
2-5.3 INPUT/OUTPUT EXPANSION. The I/O expander cards (1 and 2) are used with the I/O interface card to expand the Model 980A I/O bus as described in Paragraph 2-3. The electrical schematics of the I/O expander cards are illustrated in Figures 2-14 and 2-15. The expander cards provide buffered I/O bus signals to additional I/O backplanes at the same logic level and pins as the mainframe I/O bus ports.

The cable between the I/O interface card and the I/O expansion card permits the transfer of signals to and from the I/O interface to access the basic I/O bus. Signals are transferred from the top edge connector of the I/O interface card to I/O expander 1. Except for SYSCLK—, these signals are transferred through twisted pairs of wires that are series terminated with 82 ohms at the transmission end.

The SYSCLK— line is parallel terminated at the receiving end with 390 ohm to Vcc and ground. Provisions are included at the I/O interface to disable the 16 data input lines (DP0-15) when an expansion cable is not connected to the top edge connector of the card. All output signals to the expansion backplane from I/O expanders 1 and 2 are driven by 74H40 TTL networks.

2-5.4 INPUT/OUTPUT SIGNALS. The Model 980A I/O bus signals are listed in Table 2-3. These signals are listed as found in both the basic mainframe backplane and the expansion backplanes. Pin numbers and descriptive remarks are also included.

Expansion backplane interrupt line locations are listed in Table 2-4. As previously described, interrupt lines from each I/O port are ORed to form the I/O interrupt at the CPU. Normally, software programs identify the source of the interrupt via an RDS instruction (with the correct external register number) which transfers the interrupt line states to the CPU. Table 2-4 provides a cross listing to allow interrupt bits to be correlated to a particular I/O port connector in the expansion chassis. Thus, an interrupt that is generated by an I/O device in I/O port IO6 of the internal expansion backplane, or in I/O port 015 of the external expansion backplane, will cause bit 6 of the data word that is read by the RDS interrupt inquiry instruction to be a ONE.



BITS 13, 14, 15

BITS
9
11
12

	000	001	010	011	100	101	110	111
000	RTC PIF	PIF	TTY TTY	TTY	RTC RTC	RTC	RTC	RTC
001	RTC RTC	TTY TTY	TTY	TTY	RTC RTC	RTC	RTC	RTC
010	PTR PTR	PTP PTP	INT	PGC	PGC	PGC	PGC	PGC
011	PTR PTP	TTY	CAS	CAS	CAS	CAS	CDR	CDR
100	CDP CDP	XMD XMD	RMD RMD	ADC	ADC	ADC	ADC	ADC
101	LLP	LLP	LLP	DAC	DAC	DAC	DAC	DAC
110	EXP	EXP	EXP	DAC	DAC	DAC	DAC	DAC
111	EXP	EXP	EXP	DAC	DAC	DAC	DAC	DAC

A cross reference of register numbers, devices, and related functions is listed in Table 2-2

FIGURE 2-9. EXTERNAL REGISTER NUMBER ASSIGNMENTS

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TABLE 2-2. EXTERNAL REGISTER NUMBER ASSIGNMENTS

Register Number *	Device	Function
00	PIF – Privileged Instruction Feature RTC – Real Time Controller	Out – Set Lower Memory Boundary IN – Read RTC Status
01	PIF	Out – Set Upper Memory Boundary
02	TTY – ASR33, 700 or CRT	In – Read Data Out – Write Data
04	RTC – Real Time Controller	In – Read Time Word 1 (minutes and seconds) Out – Load Timer
07	RTC – Real Time Controller	In – Read Timer (counter register)
10	RTC – Real Time Controller	In – Read Time Word 2 (hours)
12	TTY – ASR33, 700 or CRT	In – Read Status Out – Write Control
14	RTC – Real Time Controller	In – Read Time Word 3 (days) Out – Write Control
20	PTR – Paper Tape Reader	In – Read Status Out – Write Control
21	PTR – Paper Tape Punch	In – Read Status Out – Write Control
23	INT – Interrupt Expander	In – Read Interrupt
24	PGC – Parity Generator/Checker	In – Read Parity Out – Write Even Parity
25	PGC – Parity Generator/Checker	In – Check Parity Out – Write Odd Parity
26	PGC – Parity Generator/Checker	In – Read Lock Switch
30	PTR – Paper Tape Reader PTP – Paper Tape Punch	In – Read Data Out – Write Data
32	TTY – ASR33, 700 or CRT	Disconnect Function
34	CAS – Tape Cassette System	In – Read Status (Interface) Out – Control (Interface)

TABLE 2-2. EXTERNAL REGISTER NUMBER ASSIGNMENTS (Continued)

Register Number *	Device	Function
35	CAS	In – Device Data and Control Out – Device Data and Status
36	CAS	In – Read Output Out – Bit Control
37	CDR – Card Reader	In – Read Status and Data Out – Write Control
40	CDP – Card Punch	In – Read Status Out – Write Data and Control
41	XMD – Data Set Transmit Module	In – Read Status Out – Data and Control
42	RMD – Data Set Receive Module	In – Read Data and Status Out – Control
44,45,46 54,55,56	ADC – Analog to Digital Conversion	Data Module – Same as CAS
60,61,22	LLP – Low Speed Line Printer	Data Module
64,65,66 74,75,76	DAC – Digital to Analog Conversion	Data Module Data Module
70	EXP – Interrupt Expander	In – Read Mask Out – Mask
71	EXP	Out – Unmask
72	EXP	In – Set Interrupt Out – Read Interrupt
73	EXP	Out – Reset Interrupt

* Register Number = Bits (9, 11, 12, 13, 14, 15)₈

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TABLE 2-2. STANDARD 980A EXTERNAL REGISTER/FUNCTION DETAILS

REGISTER NUMBER Bits (9-15)	DEVICE	FUNCTION
00	PIF - Privileged Instruction Feature	Out - Set lower memory boundary
01	PIF - Privileged Instruction Feature	Out - Set upper memory boundary
02	TTY1 - ASR33, Silent 700 or CRT (primary) Old Interface P/N 217394	In - Read data Out - Write data
03	TTY2 - (Secondary) Old Interface P/N 217394	In - Read data Out - Read data
04	Interval Timer	
05	Communication Module (Primary Addr.) ASR-33 KSR/ASR 733 or CRT. New Interface P/N 966637	In - Input data and Status Out - Output data and Commands
06	Communication Module (Secondary Addr.)	In - Input data and Status Out - Output data and Commands
07	Customer Use	
08	Customer Use	
0A	TTY1 - ASR-33, Silent 700 or CRT (primary) Old Interface P/N 217394	In - Read status Out - Write Command
0B	TTY2 - Secondary (Old Interface P/N 217394)	In - Read status Out - Write command
0C	Customer Use	
0D	Communication Module (Secondary Addr.)	In - Input data and Status Out - Output data and Commands
0E	Communication Module (Secondary Addr.)	In - Input data and Status Out - Output data and Commands
10	PTR - Paper Tape Reader	In - Read status Out - Write command
11	PTP - Paper Tape Punch	In - Read status Out - Write command
13	INT - Interrupt Expander Address for Expander Chassis Interrupt Expander is relative to board location	In - Read interrupt
14	Customer Use	
15	Customer Use	
16	Customer Use	
18	PTR - Paper Tape Reader PTP - Paper Tape Punch	In - Read data Out - Write data

TABLE 2-2. STANDARD 980A EXTERNAL REGISTER/FUNCTION DETAILS (Continued)

REGISTER NUMBER Bits (9-15)	DEVICE	FUNCTION
1A	TTY1 - ASR33, 700 or CRT (primary) Old Interface P/N 217394	Disconnect function
1B	TTY2 - (secondary) Old Interface P/N 217394	Disconnect function
1F	GDR - Card Reader	In - Read status and data Out - Write command
40	CDP - Card Punch	In - Read status Out - Write data and command
41	Customer Use	
42	Customer Use	
44-46	ADA - Analog to Digital Converter or Digital to Analog Converter	Data Module - (See ADA User's Manual)
4C-4E		
50	LLP - Low Speed Line Printer	Data Module - (See Line Printer User's Manual)
54	Vectored Interrupt (I/O 1&2)	Set Mask
55	Vectored Interrupt (I/O 1&2)	Clear Mask
56	Vectored Interrupt (I/O 1&2)	Set Interrupt (T)
57	Vectored Interrupt (I/O 3&4)	Set Mask
58	Vectored Interrupt (I/O 3&4)	Clear Mask
59	Vectored Interrupt (I/O 3&4)	Set Interrupt (T)

(T) Test Only